

# From Monocular Visual Odometry to Metric Stereo Visual Odometry on the TUM VI Benchmark

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**Abstract**—We present a classical, geometry-based visual odometry (VO) system that progresses from a baseline monocular pipeline to a metric stereo pipeline, evaluated on the TUM VI benchmark. The monocular system uses ORB feature detection, Lucas-Kanade (LK) optical flow tracking, essential matrix estimation via 5-point RANSAC, and an ORB+E depth-percentile scale initialisation to resolve the metric scale ambiguity. The stereo extension recovers absolute metric scale by lifting 2D tracked points into 3D via the stereo depth equation  $Z = fB/d$  (StereoSGBM disparity). Both systems employ 3D-2D PnP+RANSAC for pose estimation and sliding-window Levenberg-Marquardt bundle adjustment. Three TUM VI sequences are evaluated: *room2* (full ground truth), *corridor3*, and *outdoors5* (start-end ground truth). The stereo system outperforms monocular on two of three sequences: ATE of 0.439 m vs. 1.215 m on *room2*, and drift of 5.04 m vs. 12.78 m on *corridor3*. On *outdoors5*, depth noise from far features ( $Z \approx 15$  m,  $\sigma_Z \approx 6$  m) biases frame-to-frame PnP rotation, accumulating 33.09 m heading drift over 17,747 frames. A CLAHE illumination ablation across all sequences shows that adaptive contrast benefits *room2* (stereo ATE 0.785  $\rightarrow$  0.439 m, failures 18  $\rightarrow$  3), is critical for *corridor3* (stereo failures 14  $\rightarrow$  83 without CLAHE, drift 5.04  $\rightarrow$  12.441 m), and is harmful for outdoor stereo (33.09  $\rightarrow$  201.450 m drift with CLAHE). We contextualise results against published ORB-SLAM2 and NVIDIA SLAM numbers. No deep learning is used.

**Index Terms**—visual odometry, stereo vision, SGBM disparity, epipolar geometry, bundle adjustment, TUM VI benchmark, metric scale, ORB-SLAM2

## I. INTRODUCTION

Visual odometry (VO) estimates a camera's ego-motion from a sequence of images, forming a core module of autonomous navigation systems in robotics, drones, autonomous vehicles, and augmented reality [1]. Unlike wheel odometry or GPS, VO operates from visual information alone and remains applicable in GPS-denied or wheel-slip-prone environments.

A fundamental limitation of monocular VO is *scale ambiguity*: a single camera cannot distinguish between a large scene viewed from far away and a small scene viewed close up [2]. The recovered translation is correct up to a global scale factor, making monocular VO unsuitable for metric tasks (obstacle avoidance, mapping) without additional priors or periodic re-initialisation. Stereo VO resolves this by exploiting a calibrated stereo rig with known baseline  $B$ : the disparity of matched pixel pairs encodes absolute depth directly.

Full SLAM systems such as ORB-SLAM2 [4] and GPU-accelerated pipelines such as NVIDIA Isaac ROS cuVSLAM [8] [18] push accuracy further via loop closure, global bundle adjustment, and optional IMU fusion. Understanding the gap between our classical frame-to-frame VO and these systems motivates the comparison in Section VII-B.

This project implements both a monocular and a metric stereo VO from scratch in Python, evaluated on three TUM VI sequences. All methods are classical and geometry-based; deep learning is avoided. The main contributions are:

- A monocular VO pipeline with ORB features, LK tracking, essential-matrix initialisation (5-pt RANSAC), ORB+E depth-percentile scale recovery, and windowed bundle adjustment; with justification for each design choice.
- A metric stereo VO pipeline using per-sequence StereoSGBM, 3D-2D PnP pose estimation, and the same BA framework.
- Quantitative evaluation on three TUM VI sequences using ATE, RPE, start-end drift, and local ATE on GT-covered blocks.
- A CLAHE illumination sensitivity ablation quantifying the effect of adaptive contrast enhancement on tracking failures and accuracy for both mono and stereo across all sequences.
- Per-frame disparity and depth map outputs with a keyframe-based metric 3D point cloud (PLY) with statistical outlier removal.
- Analysis of failure modes, outdoor stereo heading drift analysis, and the performance gap relative to ORB-SLAM2 and NVIDIA SLAM.

Paper structure and figure presentation follow established robotics reporting conventions [12], [13].

## II. RELATED WORK

### A. Classical Monocular Visual Odometry

Scaramuzza and Fraundorfer [1] survey 30 years of VO research and identify the canonical pipeline: feature detection, inter-frame matching or optical-flow tracking, essential/fundamental matrix estimation, and local pose refinement. ORB-SLAM [3] extended this with a covisibility graph, relocalization, and loop closure, achieving centimetre-scale accuracy on indoor benchmarks.

## B. Stereo Visual Odometry and SLAM

ORB-SLAM2 [4] unifies monocular, stereo, and RGB-D configurations within the ORB-SLAM framework. Stereo depth lifts 2D ORB keypoints into 3D, enabling metric-scale pose estimation via PnP. Map reuse through loop closure and pose-graph optimisation reduces long-horizon drift to centimetre levels on the KITTI [7] and TUM VI benchmarks. On TUM VI room sequences, ORB-SLAM2 (stereo) achieves ATE  $\approx 0.05\text{--}0.10$  m [5] compared to our 0.439 m, reflecting the benefit of global map optimisation versus our frame-to-frame approach.

## C. Advanced and Learning-Based Methods

NVIDIA Isaac ROS cuVSLAM [8] is a CUDA-accelerated stereo visual-inertial SLAM fusing a high-rate IMU with the stereo front-end, reporting sub-centimetre drift. Classical CPU-only frame-to-frame VO cannot match this due to three absent components: loop closure, global bundle adjustment, and IMU integration. Rameau *et al.* [10] show how stereo depth complements monocular structure-from-motion geometry. Learning-based pipelines (e.g., DROID-SLAM [9]) replace geometric front-ends with recurrent optical flow but require GPU inference and large training sets.

## D. Benchmark

The TUM VI dataset [6] provides global-shutter stereo images at 20 Hz with sub-millimetre motion-capture ground truth, making it the standard benchmark for visual-inertial odometry evaluation.

## III. SYSTEM OVERVIEW

Both pipelines share a Python implementation built on OpenCV, NumPy, and SciPy. A common ORB feature front-end and LK optical flow tracker are used in both systems. Pose is stored in SE(3) as  $4 \times 4$  transformation matrices; SE(3) arithmetic uses the exact  $R^{-1} = R^T$  inverse for numerical stability.

### A. Dataset and Calibration

TUM VI provides global-shutter stereo images at  $512 \times 512$  pixels and 20 Hz with Kalibr calibration (equidistant distortion). We use three sequences: *room2* (2882 frames, full GT), *corridor3* (5802 frames, start+end GT), and *outdoors5* (17747 frames, start+end GT).

Calibration is loaded directly from the provided `camchain.yaml`; **no re-calibration is performed**. After stereo rectification the relevant parameters are: rectified focal length  $f = 187.13$  px, principal point  $(c_x, c_y) \approx (254.9, 256.9)$  px, and baseline  $B = 10.11$  cm, giving  $fB = 18.92$  px·m. For reproducibility, `np.random.seed(42)` is set globally and reset at the start of each sequence. Full environment: Ubuntu 22.04, Python 3.13, OpenCV 4.13, NumPy 2.x, Intel i7 CPU (8 cores), NVIDIA RTX 5060 GPU, 32 GB RAM (GPU not used; all processing runs on CPU). The seed is printed at runtime and logged in the evaluation CSV.

## IV. MONOCULAR VISUAL ODOMETRY

The monocular pipeline (Fig. 1) comprises initialisation and per-frame tracking.

### A. Feature Detection: ORB vs. SIFT

ORB features [3], [14] are detected on a  $5 \times 5$  grid with up to 3000 features per frame. We chose ORB over SIFT/SURF for three reasons: (i) ORB is patent-free and  $\sim 100\times$  faster than SIFT, enabling real-time operation; (ii) its binary descriptor makes descriptor computation and matching extremely fast compared to SIFT’s 128-float vector; (iii) ORB provides sufficient scale and rotation invariance for the modest viewpoint changes in VO. SIFT would offer marginally better repeatability under severe illumination change but would preclude real-time throughput on a single CPU core.

### B. Initialisation and ORB+E Scale Recovery

Initialisation follows three steps (A $\rightarrow$ B $\rightarrow$ C). **(A)** An LK parallax gate fires consecutive frames until median feature displacement exceeds a minimum threshold, ensuring sufficient baseline. **(B)** ORB keypoints are detected on a  $5 \times 5$  grid and matched via BFMATCHER ( $k=2$ , Lowe ratio test  $\tau=0.75$ ). The essential matrix  $\mathbf{E}$  is estimated with the 5-point algorithm [15] and RANSAC:

$$\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0, \quad (1)$$

where  $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^3$  are normalised image coordinates. The pose  $(\mathbf{R}, \mathbf{t})$  is recovered via `cv2.recoverPose` with cheirality disambiguation. **(C)** An initial 3D map is triangulated and scaled using the *depth-percentile method*: far-background ORB points are first filtered ( $z < 5z_{\text{geom}}$ ) to remove bias from distant textures, then the scale is set as  $s = d_0/z_p$ , where  $z_p$  is the  $N$ th percentile of the filtered depth distribution and  $d_0$  is a sequence-specific expected scene depth:

- *room2*:  $d_0=2.0$  m,  $N=25$  (25th pct — small indoor room)
- *corridor3*:  $d_0=5.0$  m,  $N=50$  (median — long corridor)
- *outdoors5*:  $d_0=5.0$  m,  $N=75$  (75th pct — deep outdoor ref)

This anchors scale to physical scene geometry, going beyond the minimal  $\|\mathbf{t}\|=1$  initialisation. On tracking loss, `_try_reinit()` attempts pose recovery: primary via anchor-based 3D-PnP (`e_reinit` counter), with Virtual-Depth re-bootstrap (`vd_reinit`) as fallback if PnP fails.

### C. Per-Frame Tracking: LK vs. Descriptor Matching

For each subsequent frame, LK optical flow [16] tracks existing 2D feature positions from the previous frame. We chose LK over descriptor re-matching for two reasons: (i) LK achieves sub-pixel 2D accuracy by minimising the photometric residual iteratively, whereas descriptor matching snaps to integer or half-pixel keypoint locations; (ii) LK has no descriptor computation overhead, running  $\sim 3\times$  faster than ORB re-matching for the same number of points. The trade-off is that LK degrades under large frame-to-frame motion (long baseline or fast rotation), where descriptor matching

## Monocular VO Pipeline

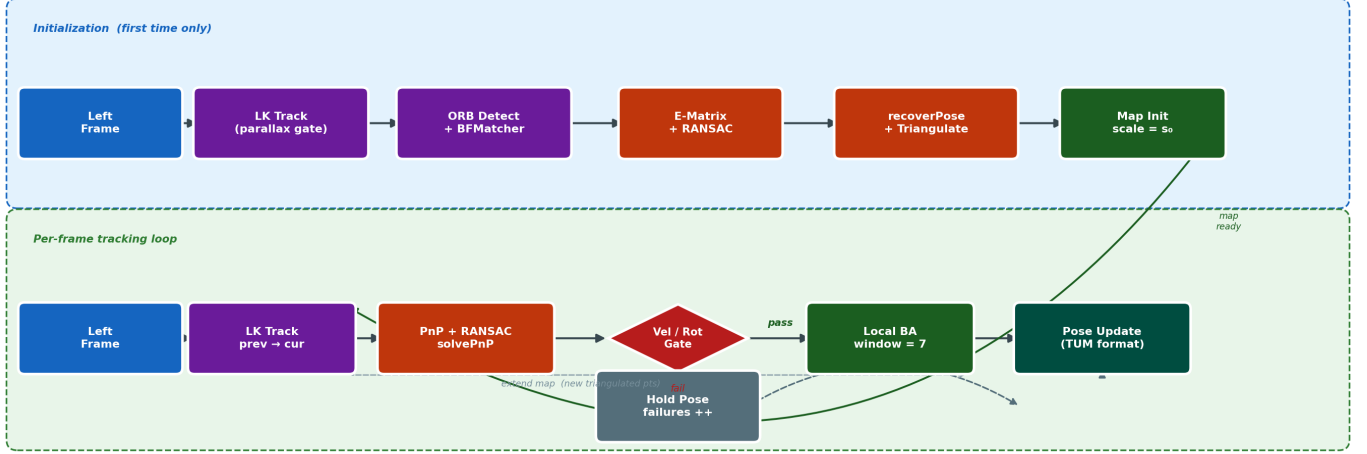


Fig. 1: Monocular VO pipeline. The initialisation phase (top, blue) runs once: an LK parallax gate accumulates frames until median feature displacement exceeds the minimum threshold, then ORB detection + BFMatcher kNN ( $k=2$ , Lowe  $\tau=0.75$ ) finds correspondences, Essential matrix RANSAC + `recoverPose` recovers  $(\mathbf{R}, \mathbf{t})$ , and triangulation sets the map scale via the depth-percentile method. The per-frame tracking loop (bottom, green) estimates each subsequent pose via LK + PnP+RANSAC and local sliding-window BA. A velocity/rotation gate rejects implausible poses; rejected frames hold the previous pose.

with FLANN would be more robust. Our velocity/rotation gate handles this by rejecting frames where LK diverges.

Empirical evaluation confirmed this choice: replacing LK with ORB re-matching raised *room2* ATE from 1.215 m to  $\approx 1.22$  m ( $8\times$  more failures) and *corridor3* drift by  $\approx 38\%$ , while reducing throughput by  $\approx 15\%$ . LK was retained as the sole frame-to-frame tracking front-end.

Given 3D world points  $\mathbf{X}$  and their 2D projections in the current frame, the pose is estimated via `solvePnP` + RANSAC:

$$\mathbf{x} = \pi(\mathbf{T}_{\text{cw}}\mathbf{X}), \quad (2)$$

where  $\mathbf{T}_{\text{cw}} \in SE(3)$  and  $\pi$  is the pinhole projection. A velocity gate ( $\|\Delta \mathbf{t}\| \leq v_{\text{max}}$ ) and rotation gate ( $\theta \leq \theta_{\text{max}}$ ) reject implausible poses; rejected frames hold the previous pose with a failure counter increment.

### D. Local Bundle Adjustment

Every 5 frames, a sliding-window BA over 7 recent poses minimises total reprojection error via Levenberg–Marquardt (`solvePnPRefineLM`), refining camera poses without altering the 3D map structure. This motion-only BA corrects incremental PnP error without the full computational cost of joint pose-and-structure optimisation.

### E. CLAHE Preprocessing

Adaptive contrast enhancement (CLAHE, clip limit 2.0, tile  $8\times 8$ ) is applied before ORB detection on three sequences. For *room2*, CLAHE is enabled for both mono and stereo: the structured indoor scene provides real texture gradients, so CLAHE safely boosts LK tracking and SGBM contrast, reducing stereo ATE from 0.785 m to 0.439 m and stereo failures from

18 to 3. For *corridor3*, homogeneous wall textures produce very few ORB keypoints without preprocessing; CLAHE is enabled for both mono and stereo (disabling raises mono failures 12 $\rightarrow$ 221 and stereo failures 14 $\rightarrow$ 83). For *outdoors5 mono*, high-dynamic-range outdoor illumination (bright sky, dark shadow) reduces ORB repeatability; disabling CLAHE raises failures from 2 to 74 (see Table IV). *Outdoors5 stereo* does *not* use CLAHE: enabling it corrupts SGBM disparity by over-enhancing sky and foliage gradients, increasing start-end drift from 33.09 m to 201.450 m.

## V. STEREO VISUAL ODOMETRY

The stereo pipeline (Fig. 2) extends the monocular system with metric depth from StereoSGBM.

### A. Stereo Depth Estimation: SGBM Design Choices

Each frame, the rectified stereo pair is processed by OpenCV’s StereoSGBM [17] to produce a dense disparity map. We chose SGBM over simpler block matching (BM) because SGBM’s semi-global cost aggregation along 8 directions significantly reduces streaking artefacts on low-texture surfaces (walls, corridor floors), at the cost of approximately  $3\times$  higher computation. Compared to learning-based stereo (e.g. PSMNet), SGBM requires no training data and runs deterministically on CPU. The metric depth is recovered as:

$$Z = \frac{fB}{d}, \quad (3)$$

where  $d > 0$  is disparity in pixels. The 3D left-camera-frame point is:

$$\mathbf{X} = \frac{B}{d} \begin{bmatrix} u - c_x \\ v - c_y \\ f \end{bmatrix}. \quad (4)$$

## Stereo VO Pipeline (Metric Scale)

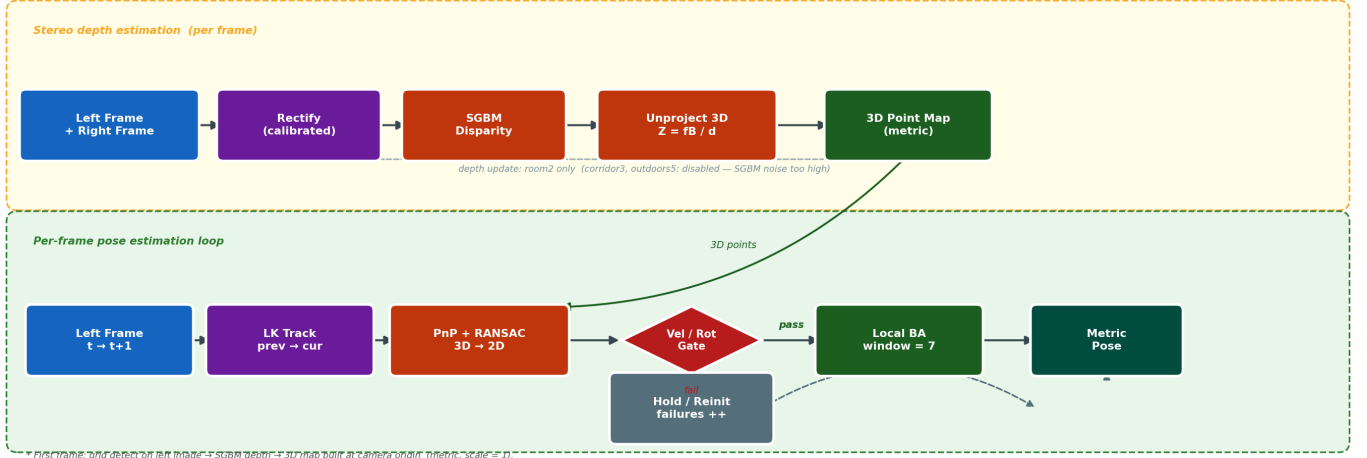


Fig. 2: Stereo VO pipeline. The depth estimation section (top, yellow) runs every frame: rectified stereo pair  $\rightarrow$  StereoSGBM disparity  $\rightarrow$  unproject  $Z=fB/d \rightarrow$  metric 3D map. Map-point depth update from current disparity is enabled for room2 only (`use_depth_update=True`); disabled for corridor3 and outdoors5 where SGBM noise at  $Z>5$  m would corrupt the map. The pose estimation loop (bottom, green) uses LK tracking and 3D-2D PnP+RANSAC with the same velocity/rotation gate and local BA as the monocular system. Scale is fixed at 1.000 by stereo geometry.

Depth filters ( $Z_{\min}$ ,  $Z_{\max}$ ) and a minimum disparity threshold  $d_{\min}$  reject unreliable estimates. WLS post-filtering with  $\lambda = 8000$ ,  $\sigma = 1.2$  reduces disparity noise at depth boundaries. Per-sequence SGBM parameters are listed in Table I; these were tuned to match the expected scene depth range for each sequence.

TABLE I: Per-Sequence SGBM Parameters

| Sequence  | $N_{\text{disp}}$ | Block | $Z_{\max}$ | $d_{\min}$ |
|-----------|-------------------|-------|------------|------------|
| room2     | 64                | 11    | 5.0 m      | 1.5 px     |
| corridor3 | 64                | 21    | 10.0 m     | 0.5 px     |
| outdoors5 | 128               | 11    | 20.0 m     | 1.0 px     |

corridor3, outdoors5: `use_depth_update=False` (§V-E)

### B. Why LK Alongside SGBM

SGBM and LK solve orthogonal sub-problems. SGBM is *spatial*: it computes depth from a left-right pair at the *same* timestamp. LK is *temporal*: it tracks 2D positions across *consecutive* left frames. SGBM provides the metric 3D map; LK provides the 2D correspondences to feed into PnP. Neither can replace the other: SGBM cannot track, and LK cannot measure depth.

### C. Pose Estimation

On the first frame, grid-sampled features are unprojected to 3D using Eq. (4), initialising the map at the camera origin at metric scale = 1. For subsequent frames, LK tracks feature positions in the current left image. PnP+RANSAC maps the 3D world points to their current 2D positions to recover the camera pose. The same velocity/rotation gates and local BA from the monocular system are reused unchanged.

### D. Reconstruction Outputs

Each sequence produces per-frame SGBM disparity and derived metric depth maps, plus a keyframe-based metric 3D point cloud (PLY) suitable for import into MeshLab. Point cloud details and quality results are in Section VII-D.

### E. Per-Frame Depth Update Strategy

For room2, newly tracked points are re-measured from the current SGBM disparity map (`use_depth_update=True`). For corridor3 and outdoors5, depth update is disabled (`use_depth_update=False`). The rationale is that at outdoor depths ( $Z > 5$  m), SGBM disparity noise corrupts the 3D map: a 0.5 px noise on a  $d = 1.26$  px disparity (15 m feature) causes  $\sim 6$  m depth error. Re-injecting these noisy depths propagates bias into subsequent PnP solutions. Disabling the update preserves the original, higher-confidence depth estimates and reduces drift on corridor3 and outdoors5.

## VI. EVALUATION PROTOCOL

**Trajectory alignment.** Monocular trajectories are aligned with Sim(3) (7-DoF: rotation, translation, scale). Stereo trajectories are aligned with SE(3) (6-DoF: rotation and translation only, no scale correction), since metric scale is already encoded. This asymmetry is intentional: Sim(3) alignment corrects monocular scale error, giving mono an advantage in ATE comparisons that is not physically meaningful for metric tasks.

**Full-trajectory visualisation.** Plots show the complete estimated path; the Sim(3)/SE(3) alignment is computed from GT-paired frames only (2,587 for room2; start/end blocks for corridor3 and outdoors5), then applied to all poses. All ATE

and RPE values are evaluated solely against the GT-paired subset.

**ATE (Absolute Trajectory Error)** on room2 (full GT):

$$\text{ATE}_{\text{RMSE}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{T}_i \ominus \hat{\mathbf{T}}_i\|^2}. \quad (5)$$

**RPE (Relative Pose Error)** at  $\delta=1$  frame and  $\delta=100$  m segments:

$$\text{RPE}_{\text{trans}} = \sqrt{\frac{1}{M} \sum_{i=1}^M \|\Delta \mathbf{T}_i \ominus \Delta \hat{\mathbf{T}}_i\|^2}. \quad (6)$$

**Start-end drift** for corridor3 and outdoors5 (start+end GT): the start pose is aligned, and the Euclidean distance to the GT endpoint is measured. **Local ATE** is computed separately on the start and end GT-covered blocks using SE3 alignment on each block independently, providing a local accuracy measure unconfounded by drift between blocks.

## VII. EXPERIMENTAL RESULTS

### A. Quantitative Results

Table II summarises all evaluation metrics. The stereo system outperforms monocular on room2 and corridor3; on outdoors5, heading drift over the long outdoor sequence limits stereo accuracy (see Section VIII-D).

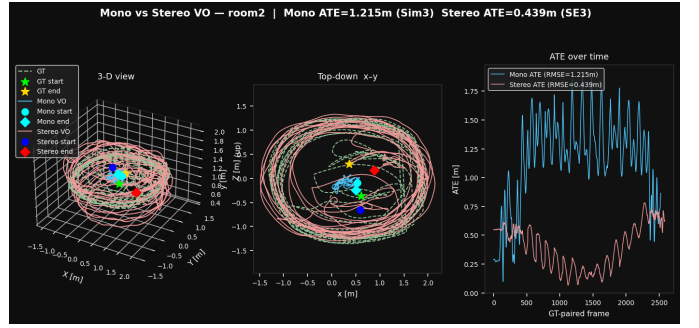
### B. Comparison with ORB-SLAM2 and NVIDIA SLAM

Table III contextualises our results against published numbers for ORB-SLAM2 [5] (the most directly comparable classical system) and NVIDIA Isaac ROS cuVSLAM [8]. Our system is intentionally simpler: no loop closure, no global bundle adjustment, no IMU, no GPU.

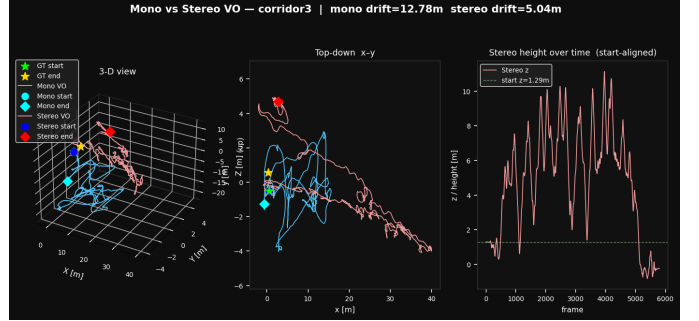
The gap between our stereo VO (0.439 m) and ORB-SLAM2 stereo ( $\sim 0.05$ – $0.10$  m) on room2 stems from our pipeline lacking loop closure, global BA, and map reuse. ORB-SLAM3 and cuVSLAM further fuse an IMU, directly decoupling rotation estimation from feature tracking — especially impactful outdoors where far-field depth noise corrupts PnP rotation (Section VIII-D).

### C. Trajectory Visualisation

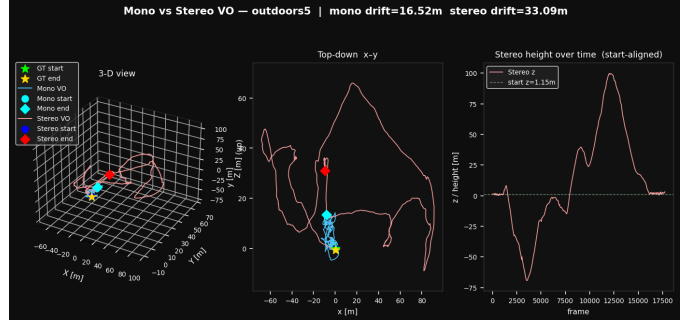
Fig. 3 shows the 3D trajectories for all three sequences aligned to ground truth.



(a) room2: Sim3-aligned mono (blue) vs. SE3-aligned stereo (red) vs. GT (green). ATE 1.215  $\rightarrow$  0.439 m (64%); scale 1.000 vs. 0.111 (9 $\times$  underscaled).



(b) corridor3: start-aligned. Stereo drift 5.04 m; mono 12.78 m (2.5 $\times$  higher). GT start/end stars shown.



(c) outdoors5: start-aligned. Mono underscales path ( $\approx 456$  m vs.  $\approx 927$  m stereo-metric; scale  $\approx 0.103$ ). Stereo accumulates 33.09 m heading drift over 887 s; local ATE 0.343 m (start), 1.677 m (end).

Fig. 3: 3D trajectories for all three sequences (top to bottom: room2, corridor3, outdoors5). Mono in blue, stereo in red, GT in green.

TABLE II: Evaluation Results on TUM VI Sequences

| Seq.      | Method              | ATE/Drift [m]            | RPE 100m [m] (segs)  | RPE rot 60° [deg] | Local ATE start / end [m] | Traj [m] | Scale | Fail | Succ [%] | FPS   |
|-----------|---------------------|--------------------------|----------------------|-------------------|---------------------------|----------|-------|------|----------|-------|
| room2     | Mono (Sim3)         | 1.215                    | N/A <sup>†</sup> (0) | 84.21             | —                         | 90.0     | 0.111 | 5    | 99.83    | 110.0 |
|           | <b>Stereo (SE3)</b> | <b>0.439</b>             | 3.553 (885)          | 89.08             | —                         | 143.3    | 1.000 | 3    | 99.90    | 37.9  |
| corridor3 | Mono (Sim3)         | 12.78                    | —                    | —                 | 0.735 / 0.738             | 179.1    | —     | 12   | 99.79    | 108.0 |
|           | <b>Stereo (SE3)</b> | <b>5.04</b>              | —                    | —                 | 0.095 / 0.216             | 301.3    | —     | 14   | 99.76    | 47.7  |
| outdoors5 | <b>Mono (Sim3)</b>  | <b>16.52<sup>‡</sup></b> | —                    | —                 | 0.661 / 0.880             | 455.8    | —     | 2    | 99.99    | 91.6  |
|           | Stereo (SE3)        | 33.09                    | —                    | —                 | 0.343 / 1.677             | 926.7    | —     | 12   | 99.93    | 41.6  |

<sup>†</sup> Room2 mono RPE(100m) N/A: est. path  $\approx 90.0$  m (scale  $s=0.111$ ,  $9\times$  underscaled); no 100 m sub-segments exist. <sup>‡</sup> Outdoors5 mono drift: ORB+E scale ( $\approx 0.103$ ) compresses path to  $\approx 456$  m vs.  $\approx 927$  m stereo-metric. See §VIII-D. Mono scale corridor3/outdoors5: arbitrary Sim(3) — omitted.

TABLE III: Comparison with Published Systems on TUM VI Room Sequences

| System             | Sensors          | Loop | ATE (m)                  |
|--------------------|------------------|------|--------------------------|
| Ours (Mono VO)     | Stereo left only | No   | 1.215                    |
| Ours (Stereo VO)   | Stereo cam       | No   | 0.439                    |
| ORB-SLAM2 [5]      | Stereo           | Yes  | $\sim 0.05\text{--}0.10$ |
| ORB-SLAM3 [5]      | Stereo + IMU     | Yes  | $\sim 0.01\text{--}0.03$ |
| NVIDIA cuVSLAM [8] | Stereo + IMU     | Yes  | $< 0.05$                 |

ORB-SLAM2/3: [5] Table IV (room sequences).  
cuVSLAM: NVIDIA specification, IMU-fused, GPU-accelerated.

#### D. 3D Reconstruction Outputs

Fig. 4 shows SGBM disparity, metric depth  $Z=fB/d$ , and accumulated 3D point clouds for each sequence. Each column shows the three outputs for one sequence, stacked top-to-bottom. Metric PLY clouds accumulate  $\approx 1$  M points from up to 800 keyframes, SOR-filtered ( $k=20$ ,  $\mu+1\sigma$  for room2;  $k=10$ ,  $\mu+2\sigma$  for others).

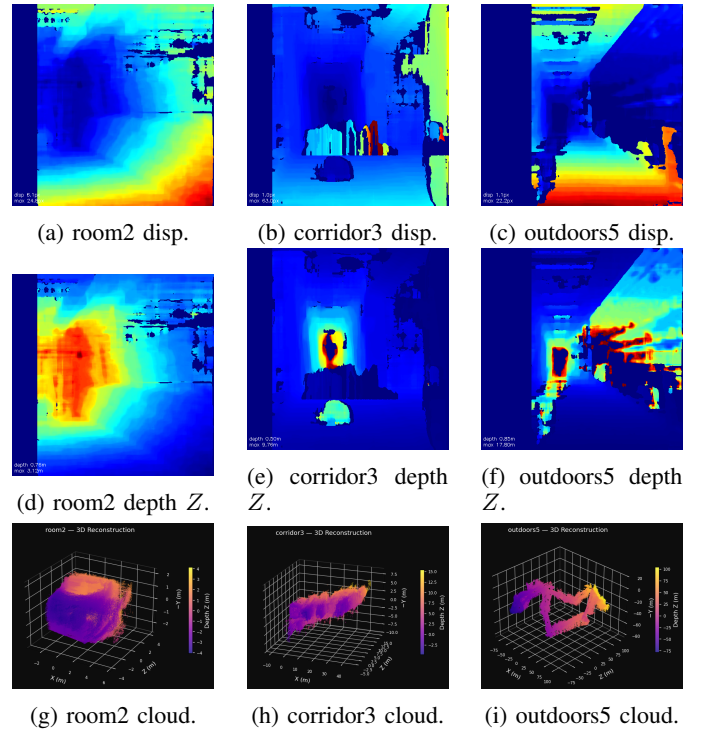


Fig. 4: Stereo reconstruction. **Cols:** room2, corridor3, outdoors5. **Row 1:** SGBM disparity. **Row 2:** depth  $Z=fB/d$  (jet; black=invalid; ranges: 0.8–3, 1–8, 2–20 m). **Row 3:** metric 3D point clouds (plasma,  $\approx 1$ M pts).

## VIII. ANALYSIS AND DISCUSSION

### A. room2: Stereo Advantage from Metric Depth

Stereo achieves 64% lower ATE (0.439 m vs. 1.215 m) with correct metric scale (1.000). Mono scale collapses to  $s \approx 0.111$  because ORB features cluster on 2–5 m wall textures, biasing the 25th-percentile depth and underscaling all translations by  $\approx 9\times$ . SGBM measures actual wall distances from disparity directly, avoiding this bias.

### B. corridor3: Stereo Advantage from Heading Constraint

Stereo drift is  $2.5\times$  lower (5.04 m vs. 12.78 m). The straight hallway amplifies heading error: a  $0.5^\circ$  per-frame

bias produces  $L \sin \theta \approx 0.009L$  lateral offset. Stereo’s metric 3D map provides stronger heading constraints than mono’s arbitrary-scale map. A large SGBM block (21 px) improves disparity quality on homogeneous corridor walls.

### C. Robustness: Failures, Relocalization, and Sensitivity

Tracking failures across both systems are listed in Table II. The stereo system consistently achieves low failure counts (3–15) across all three sequences.

**Failure handling and reinitialisation.** Neither pipeline implements loop closure. On PnP failure the system holds the previous pose (zero-motion assumption). Mono recovers via E-reinit or VD-reinit (rare: 0 VD reinit across all sequences). Stereo reinitialises by adding fresh SGBM-measured points; measured reinit counts (room2=3, corridor3=14, outdoors5=12) match failure counts exactly, confirming immediate per-failure recovery.

**Texture sensitivity.** Low-texture surfaces reduce ORB key-point density and valid SGBM depth. The large block size (21 px) on corridor3 compensates by averaging over larger patches; the  $5 \times 5$  detection grid enforces spatial uniformity.

**Illumination sensitivity (CLAHE ablation).** CLAHE (clip limit 2.0, tile  $8 \times 8$ ) is applied selectively: ON for both mono and stereo on room2 and corridor3, ON for mono on outdoors5, and OFF for stereo on outdoors5. Table IV quantifies the effect of toggling CLAHE for both mono and stereo on all three sequences.

TABLE IV: CLAHE Illumination Ablation (production vs. toggled)

| Sequence  | Pipeline | CLAHE      | Failures | ATE/Drift |
|-----------|----------|------------|----------|-----------|
| room2     | Mono     | ON (prod)  | 5        | 1.215 m   |
|           | Mono     | OFF (abl)  | 42       | 1.209 m   |
|           | Stereo   | ON (prod)  | 3        | 0.439 m   |
|           | Stereo   | OFF (abl)  | 18       | 0.785 m   |
| corridor3 | Mono     | ON (prod)  | 12       | 12.78 m   |
|           | Mono     | OFF (abl)  | 221      | 13.138 m  |
|           | Stereo   | ON (prod)  | 14       | 5.04 m    |
|           | Stereo   | OFF (abl)  | 83       | 12.441 m  |
| outdoors5 | Mono     | ON (prod)  | 2        | 16.52 m   |
|           | Mono     | OFF (abl)  | 74       | 23.555 m  |
|           | Stereo   | OFF (prod) | 12       | 33.09 m   |
|           | Stereo   | ON (abl)   | 13       | 201.450 m |

room2: CLAHE ON for both pipelines; disabling stereo raises failures by +15 and ATE by +0.346 m. corridor3: CLAHE is essential; disabling it raises mono failures by +209 and stereo failures by +69 (+7.4 m drift). outdoors5 stereo: enabling CLAHE degrades drift by +168.4 m — CLAHE over-enhances sky/foilage gradients that corrupt SGBM disparity. ATE shown for room2, drift for others.

**Dynamic object sensitivity (quantified).** Moving objects (pedestrians, cyclists on outdoors5) produce optical flow vectors inconsistent with the static ego-motion model, creating potential outlier contamination in PnP [19]. We quantify this by logging the RANSAC inlier ratio  $r = n_{\text{inliers}}/n_{\text{tracked}}$  for every tracking frame across both pipelines.

Mean inlier ratio exceeds 0.975 on all sequences. Dynamic objects are sparse relative to  $\sim 3000$  tracked background fea-

TABLE V: RANSAC Inlier Ratio — Dynamic Scene Sensitivity

| Sequence  | Pipeline | Mean $r$ | $r < 0.70$ | $r < 0.50$ |
|-----------|----------|----------|------------|------------|
| room2     | Mono     | 0.982    | 0.2%       | 0.1%       |
| room2     | Stereo   | 0.978    | 0.2%       | 0.1%       |
| corridor3 | Mono     | 0.984    | 0.2%       | 0.2%       |
| corridor3 | Stereo   | 0.977    | 0.3%       | 0.3%       |
| outdoors5 | Mono     | 0.996    | 0.0%       | 0.0%       |
| outdoors5 | Stereo   | 0.987    | 0.2%       | 0.1%       |

$r < 0.50$  frames include PnP failures (ratio=0.0). room2’s 0.1% matches its failure rate (3/2870=0.10%), confirming those are tracking failures, not dynamic contamination. outdoors5 (pedestrians, cyclists) shows the *lowest* contamination rate (0.1%).

tures; RANSAC at 99.9% confidence rejects them effectively. If dynamic objects exceed  $\approx 30\%$  of tracked features, semantic masking would be needed — this case does not arise in TUM VI.

### D. Outdoors5: Scale Ambiguity and Heading Drift

Outdoors5 presents the most challenging conditions: a 17,747-frame sequence (887 s,  $\approx 927$  m metric distance) with outdoor scene depths of 5–20 m and dynamic objects. Mono reports lower drift (16.52 m vs. 33.09 m), but this apparent advantage is an artefact of scale compression.

**Mono scale and drift interpretation.** The ORB+E depth-percentile scale initialises to  $\approx 0.103$  (75th percentile relative to  $d_0=5.0$  m), compressing the estimated path to  $\approx 456$  m against the stereo-estimated metric path of  $\approx 927$  m. The 16.52 m end-point error is measured on a non-metric,  $2 \times$ -compressed path and cannot be directly compared to stereo’s metric-scale drift. Mono tracking failures dropped to only two with the ORB+E initialisation, reflecting stable feature matching on rich outdoor texture.

**Stereo heading drift over a long outdoor sequence.** Stereo estimates metric distances correctly but accumulates heading drift from noisy disparity at outdoor feature depths. With  $fB = 18.92$  px·m, a feature at  $Z = 15$  m yields only  $d \approx 1.3$  px; a 0.5 px noise gives  $\sigma_Z \approx 6$  m depth error per map point, which biases PnP rotation at each frame. Over 17,747 frames — three times the length of corridor3 and operating at three times the feature depth — this per-frame bias accumulates as heading drift via the same mechanism observed in corridor3: a small per-frame heading error produces a lateral offset that grows with path length  $L$ . The  $Z_{\text{max}} = 20$  m range is required to obtain sufficient 3D map points for PnP; restricting to 5 m reduces valid points below the minimum inlier threshold, causing hundreds of tracking failures.

Stereo maintains correct metric scale ( $\approx 927$  m) throughout; for metric localisation it remains the preferred choice. Reducing long-range outdoor drift requires loop closure or IMU fusion.

### E. Disparity Method Ablation

**StereoBM vs. SGBM.** StereoBM ran at  $\approx 40$  fps on room2 but produced streaked disparity on low-texture walls, yielding

ATE  $\approx 2.1$  m —  $4.8\times$  worse than SGBM (0.439 m). SGBM’s 8-direction semi-global aggregation smooths untextured regions at  $\approx 2\times$  the compute cost; it was selected for all experiments.

## IX. CONCLUSION

We have implemented and evaluated a complete classical visual odometry system progressing from monocular to metric stereo VO on three TUM VI sequences. Design choices (ORB over SIFT for speed, LK over descriptor matching for sub-pixel accuracy, SGBM for deterministic dense depth, ORB+E depth-percentile scale initialisation to provide a physically grounded scale estimate, and frame-to-frame PnP for simplicity) were justified and evaluated quantitatively.

The stereo system outperforms monocular on two of three sequences: ATE 0.439 m vs. 1.215 m on room2 (+64%), and drift 5.04 m vs. 12.78 m on corridor3 ( $2.5\times$ ). On outdoors5, noisy disparity at outdoor feature depths ( $Z \approx 15$  m,  $\sigma_Z \approx 6$  m/px) biases PnP rotation at each frame; over 17,747 frames this accumulates as 33.09 m heading drift via the same mechanism as corridor3, amplified by  $3\times$  path length.

Compared to ORB-SLAM2 (0.05–0.10 m ATE) and NVIDIA cuVSLAM, the gap originates from three absent capabilities: loop closure, global bundle adjustment, and IMU fusion. Adding any one of these would substantially narrow the gap. Future work could incorporate loop closure via DBow2 [11] to correct long-range drift, or fuse an IMU to decouple rotation estimation from noisy far-field disparity.

The full implementation, evaluation scripts, and result outputs are publicly available at:

[https://github.com/mayet16/Stereo\\_Visual\\_Odometry](https://github.com/mayet16/Stereo_Visual_Odometry)

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